

Assignments 1-3 Reference Guide

(Video-Aligned)

Course: Agricultural Data Systems

Instructor: Clayton Young

What This Is

This reference guide gives you exactly what to submit for Assignments 1-3.

Important: Assignment numbers do not have to match class numbers.

Assignment 1 - Setup and Skill-Run Evidence

Required setup links

- Template repo branch:
<https://github.com/SuperiorByteWorks-LLC/agent-project/tree/feat/agri-skills-for-agent-project>
- Skills source: <https://github.com/borealBytes/ag-skills/tree/skills-content>

OpenCode import prompt (paste as-is)

```
Import skills from  
https://github.com/borealBytes/ag-skills/tree/skills-content into this  
current project and make them available for OpenCode to use.
```

What you are proving

- You can run the class workflow and generate starter outputs with skills.
- You can inspect files and explain what they are.

Submit

Item	Required?	Notes
Branch/repo link	Yes	Assignment 1 branch
Run screenshots	Yes	Environment + successful skill run

Assignment 2 - Data Bundle and Provenance Pipeline

What you are proving

- You can build a multi-dataset bundle with skills.
- You can document full provenance for generated outputs.

Dataset coverage target

- Boundaries + soil + weather + CDL
- CDL years: **more than 3** (4+ is acceptable)

Submit

Item	Required?	Notes
Branch/repo link	Yes	Assignment 2 branch
Run screenshots	Yes	Show successful skill runs

Assignment 3 - EDA Report and Provenance Evidence

What you are proving

- You can produce a professional markdown EDA deliverable with inline visuals.
- You can keep the work reproducible with script provenance.

Scope guardrail

- This is **EDA**, not a full advanced analysis project.
- Show representative summaries and data quality observations.

Submit

Item	Required?	Notes
Branch/repo link	Yes	Assignment 3 branch
Run screenshots	Yes	Show successful EDA run output

Data Authenticity Note (Assignments 2-3)

Some current skill outputs may still show synthetic/hallucinated behavior.

- You are expected to inspect outputs and flag suspicious artifacts.
- If you run later with updated skills and outputs are fully real, that is acceptable.

Quick Submission Checklist

- ☐ Branch/repo link included in submission
- ☐ Run evidence screenshots included in submission
- ☐ Optional extras are kept in repo and mentioned in Google Classroom note

Optional Extra Evidence

If you add extra evidence (summary notes, provenance notes, extra visuals), keep it in your repo and add a short note in Google Classroom saying where it is.

For easier instructor review, keep assignment artifacts in root data/ and follow data/README.md.

Extra Insights from Class Demo (Optional but Helpful)

Planning first = faster execution

- Spend more time in planning mode than you think.
- Read assumptions before approving execution.
- A strong plan lets the agent run longer with fewer interruptions.

AI workflow habits that reduce mistakes

- Run one write session at a time per branch.
- Let the agent self-correct minor errors before interrupting.
- Ask for complete provenance (scripts + outputs), not outputs alone.

Git/branch habits used in demo

- Never do assignment work directly on main.
- Use feature branches per assignment and push frequently.
- Keep commit messages descriptive so review history is clear.

VS Code / Codespaces operating tips

- Codespaces is safer when you are learning: easy reset, less local risk.

- Use side-by-side panel layout to view code and rendered markdown together.
- Keep one focused terminal/session for the active assignment run.

Data realism and scope control

- Prefer real field/API data over synthetic placeholders.
- Start with a small sample to verify pipeline, then scale up.
- Keep Assignment 3 in EDA scope: clear summaries, visuals, and data-quality notes.

Model usage note

- Free models can complete this work; they may just need more iterations.
- Good prompts and planning quality matter more than model price for these assignments.

If you are unsure whether something is required, follow this sheet first.